

Alpha, Beta, and the Geometry of Diversification

Stage 1 - Within equities

Formula: $\sigma_p^2 = \sigma^2 * (\rho + (1-\rho)/N)$

Equity beta floor: 10.95%

90% of available diversification captured by about N=14

5 bps marginal cutoff reached by about N=16

Stage 2 - Across macro sleeves

Mode: rose_proxies

Covariance method: pairwise_overlap_stabilized

Five factors explain 93.6% of sleeve variance

Six factors explain 96.0% of sleeve variance

Selected factor ERC count: 6

Stage 3 - Beta fund basket and optimized beta portfolios

Fund count: 20

Accessible comparison window: 2022-01-05 to 2026-04-17

Flagship-fund overlap window: 2025-03-07 to 2026-04-17

Most SPY-like fund: NTSX

Most distinct fund: AQMIX

Closest to constructed 3x: RPAR

Best factor-balanced fund: RSSY

Most diversifying to SPY: QDSIX

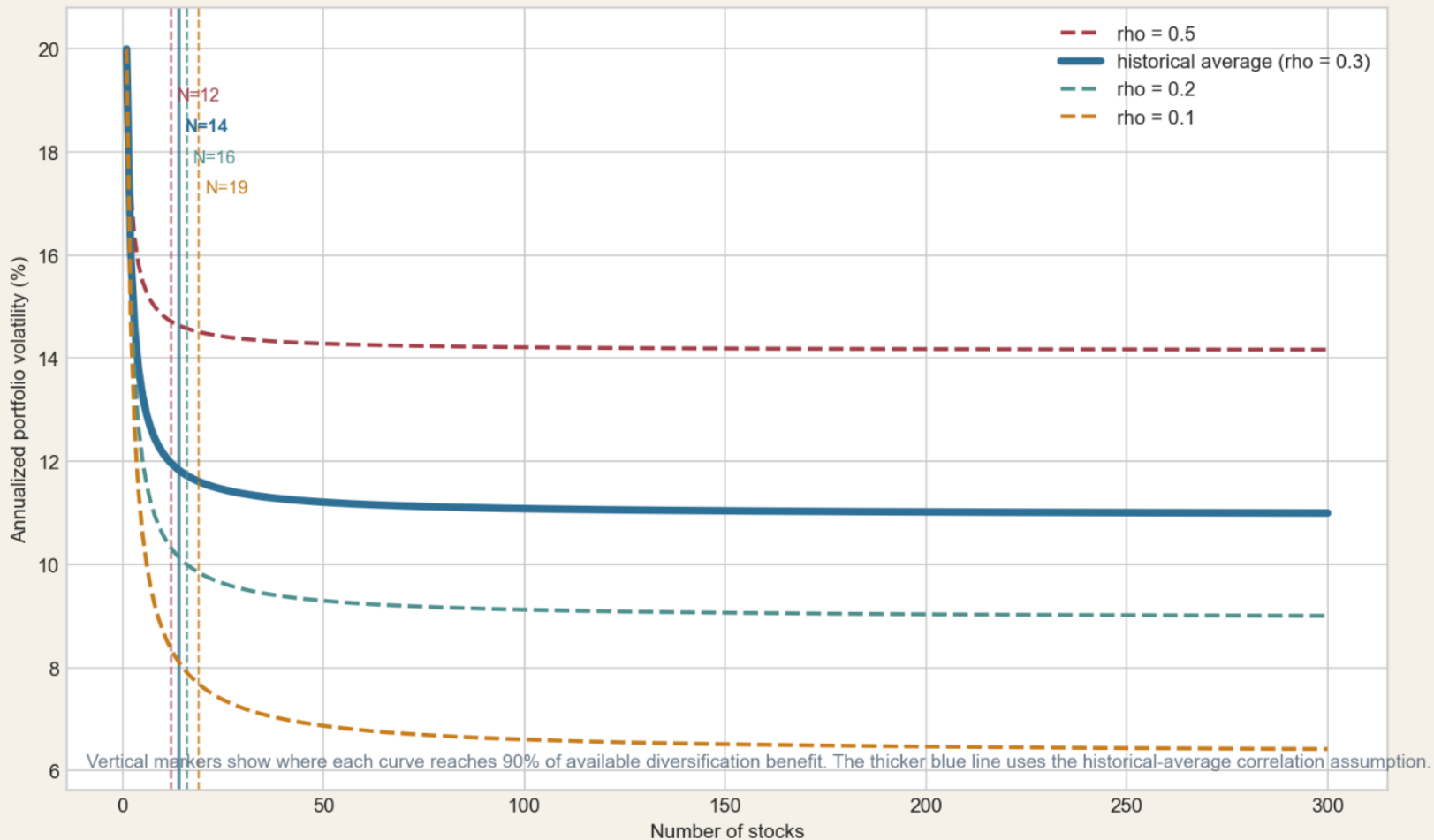
Best fee-efficient diversifier: WTMF

Rolling top-5 dynamic portfolio: 175 monthly observations, avg leverage 3.00x

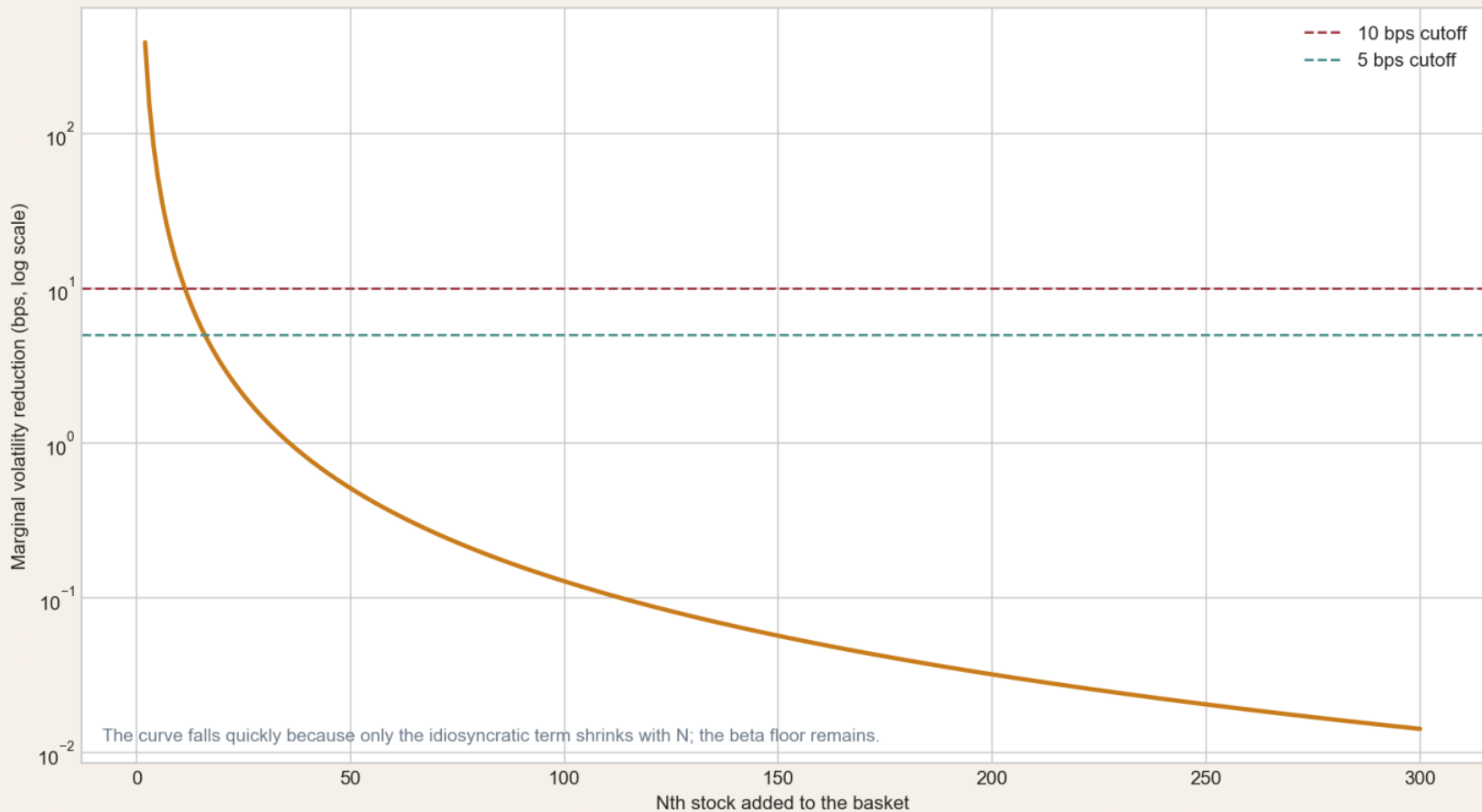
Research anchors

- Bridgewater - The All Weather Story
- Meir Statman (1987) - How Many Stocks Make a Diversified Portfolio?
- AQR - Understanding Risk Parity
- Ang, Bass, Gladstone et al. - Total Portfolio Factor, Not Just Asset, Allocation
- MSCI - Multi-Asset Class Factor Model Tier 1
- Attilio Meucci - Re-Defining and Managing Diversification
- James, Menzies, Gottwald (2022) - On financial market correlation structures and diversification benefits across and within equity sectors

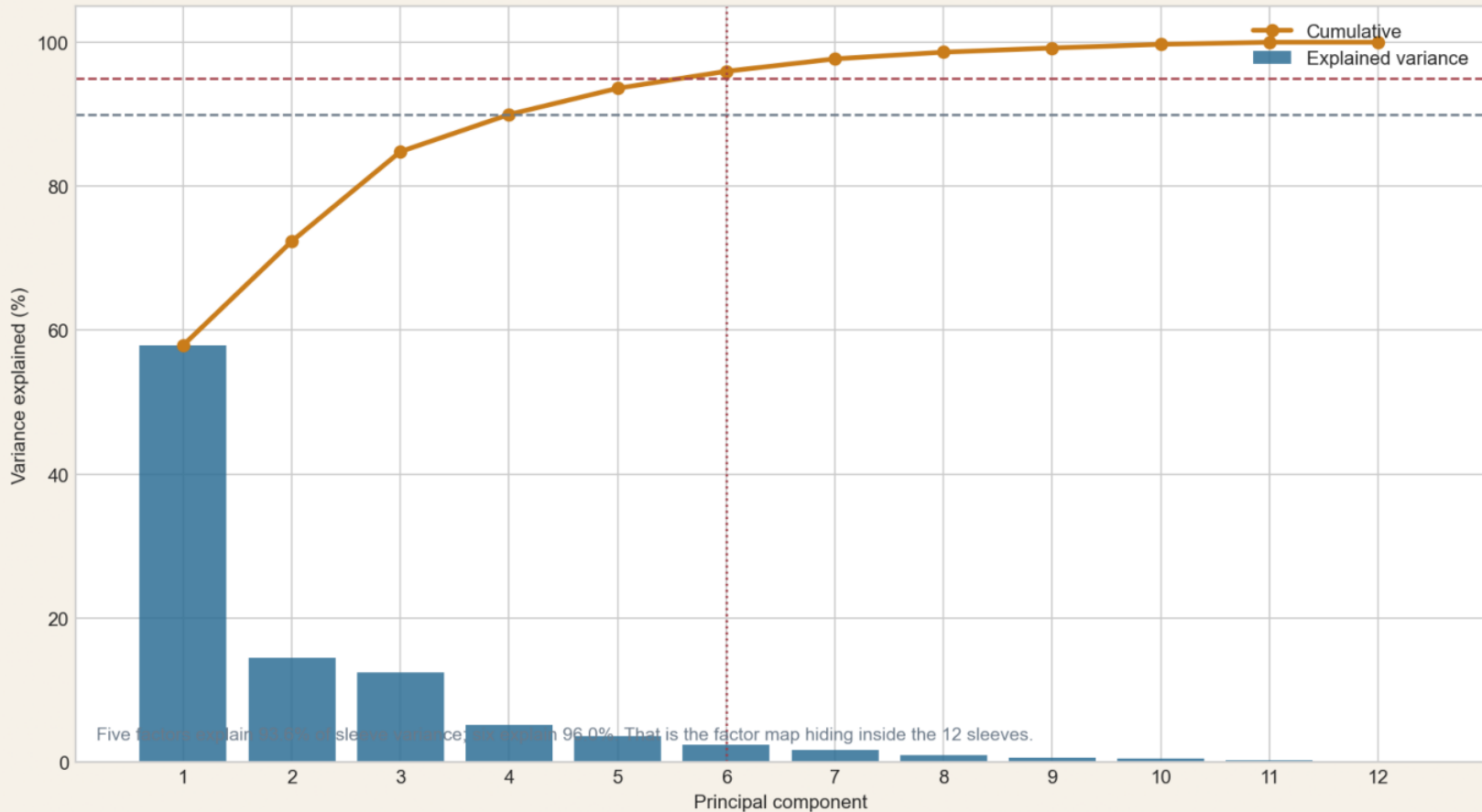
Stage 1: Absolute Volatility vs Stock Count Across Correlation Regimes



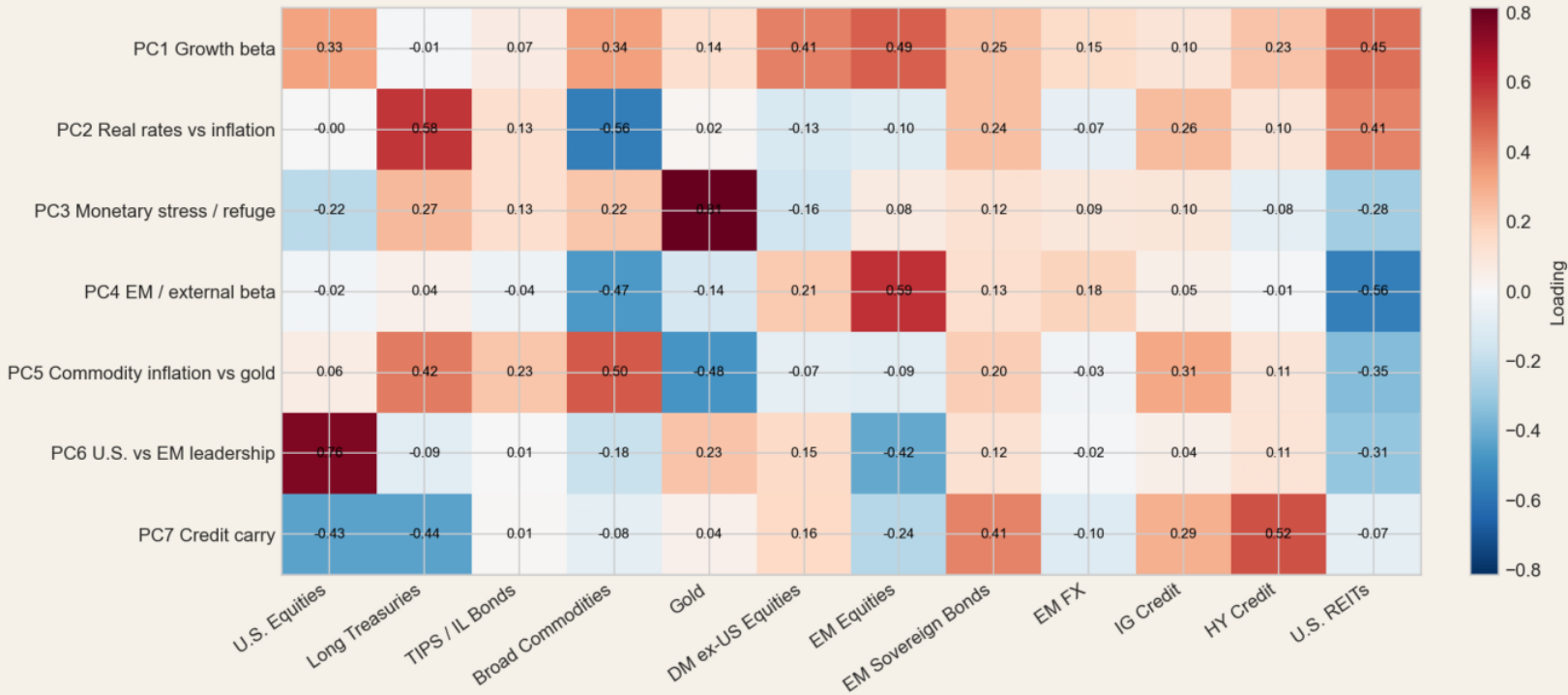
Stage 1: Marginal Benefit of the Next Stock



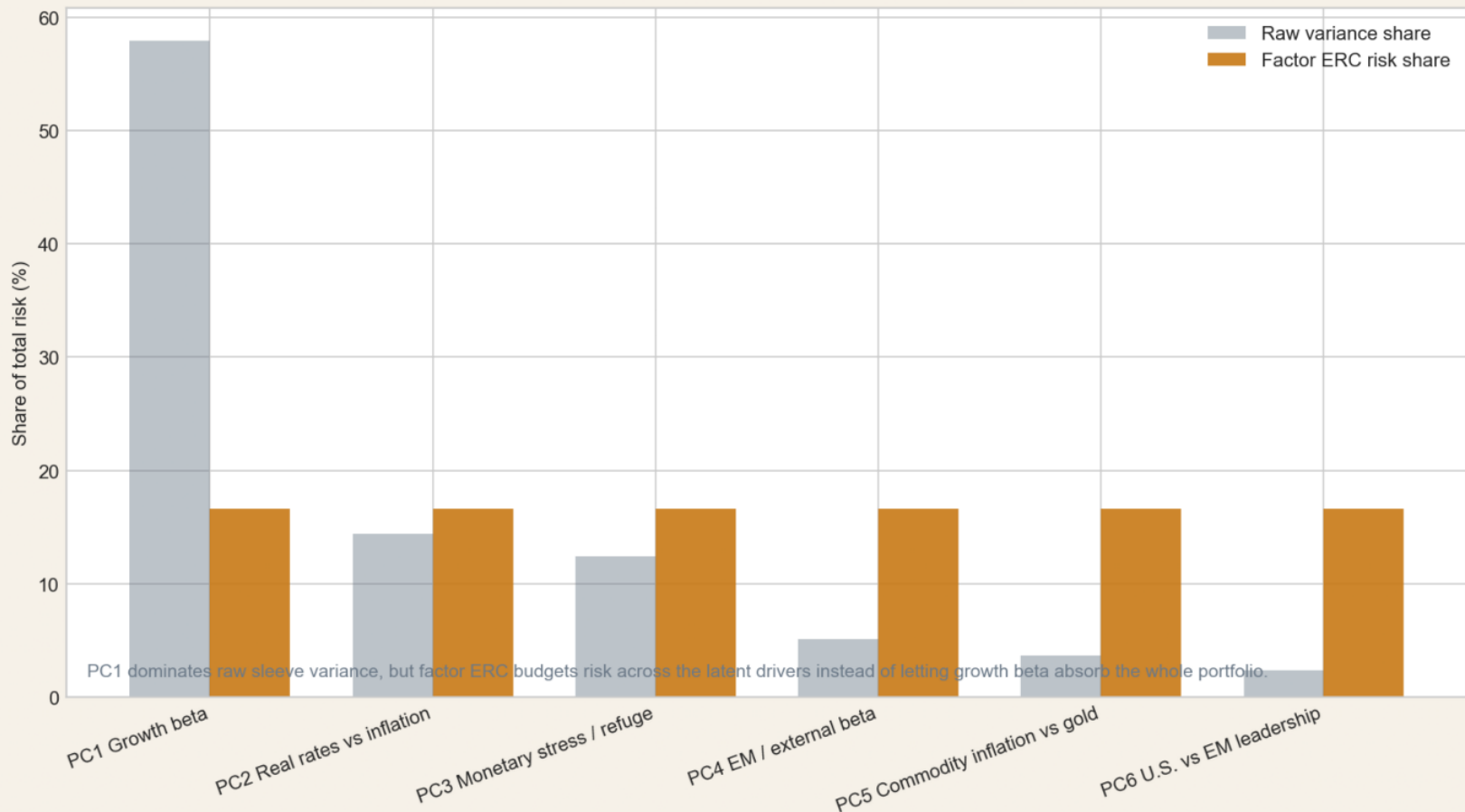
Stage 2: Covariance PCA Scree Plot



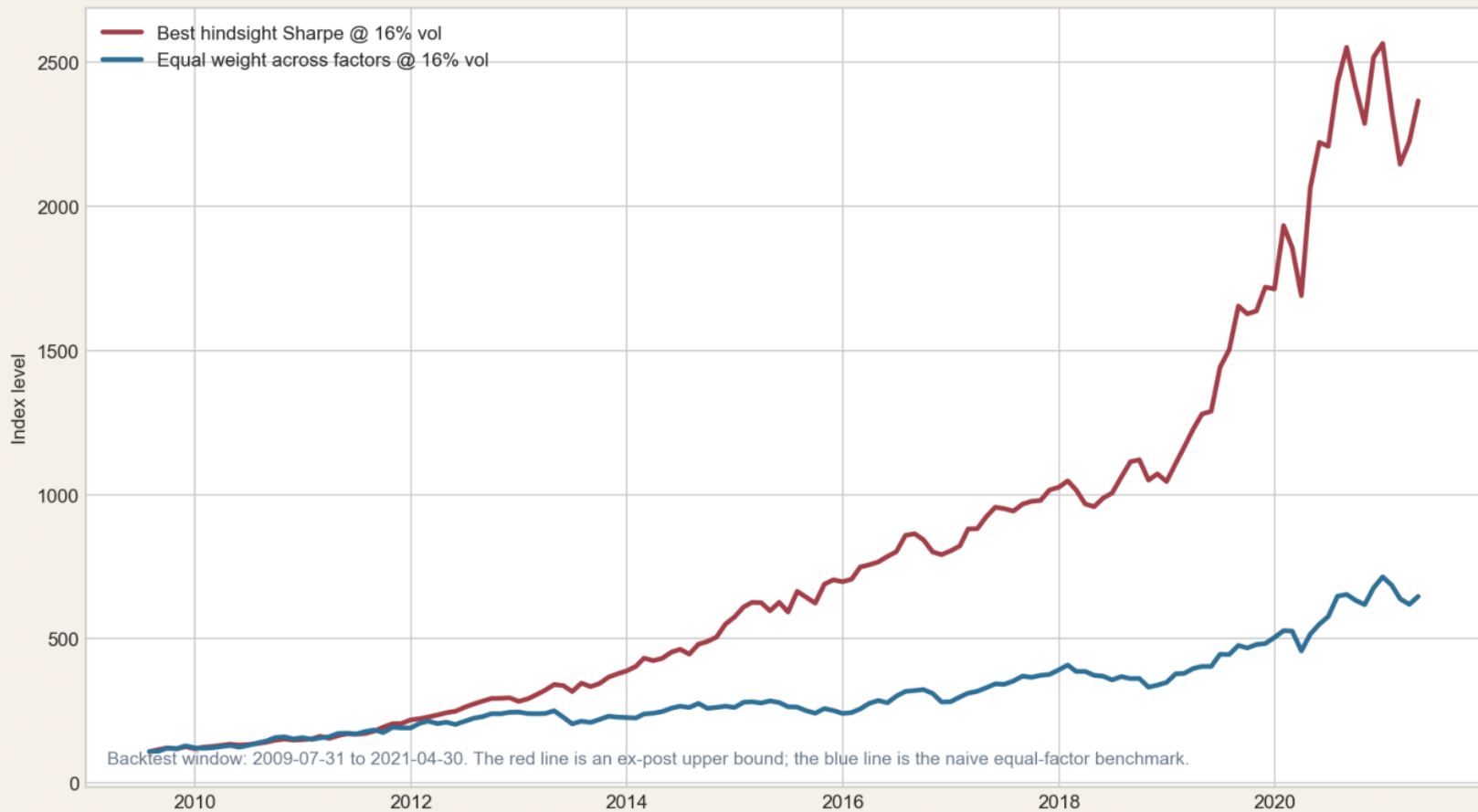
Stage 2: First Seven Principal Components And Sleeve Loadings



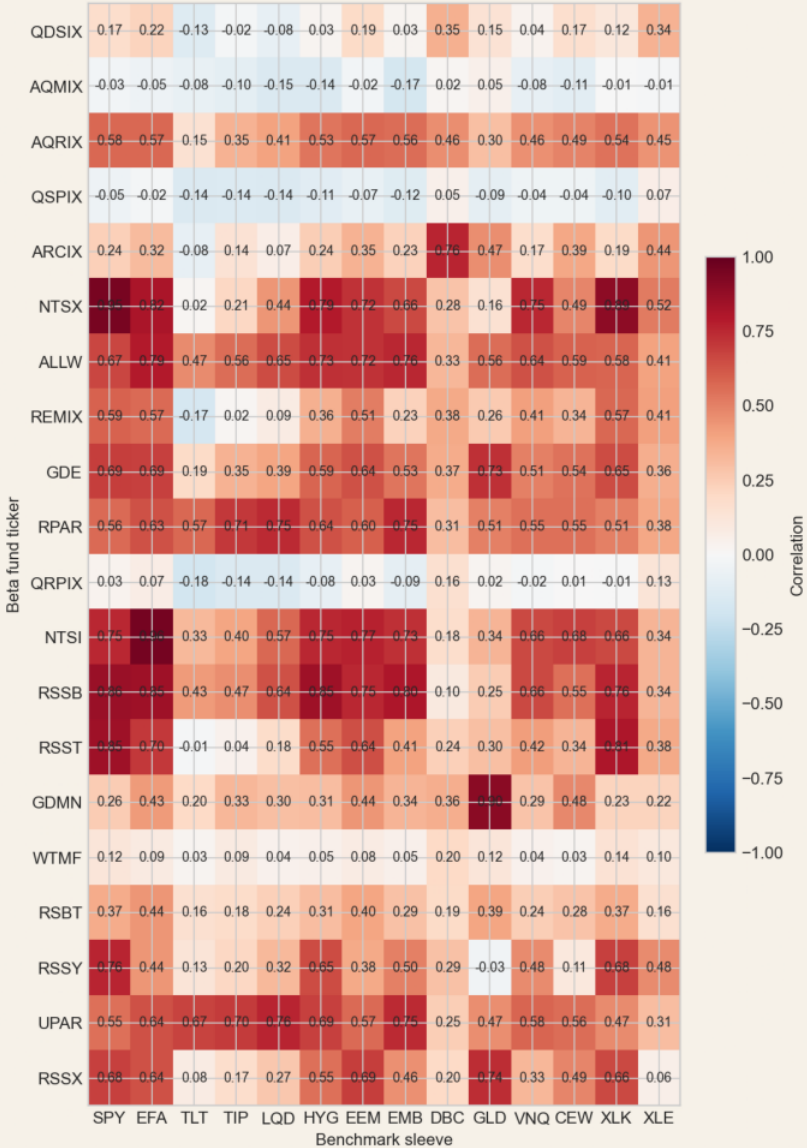
Stage 2: Equal Risk Contribution Across The Principal Factors



Stage 2: 16% Vol-Targeted Factor Portfolios

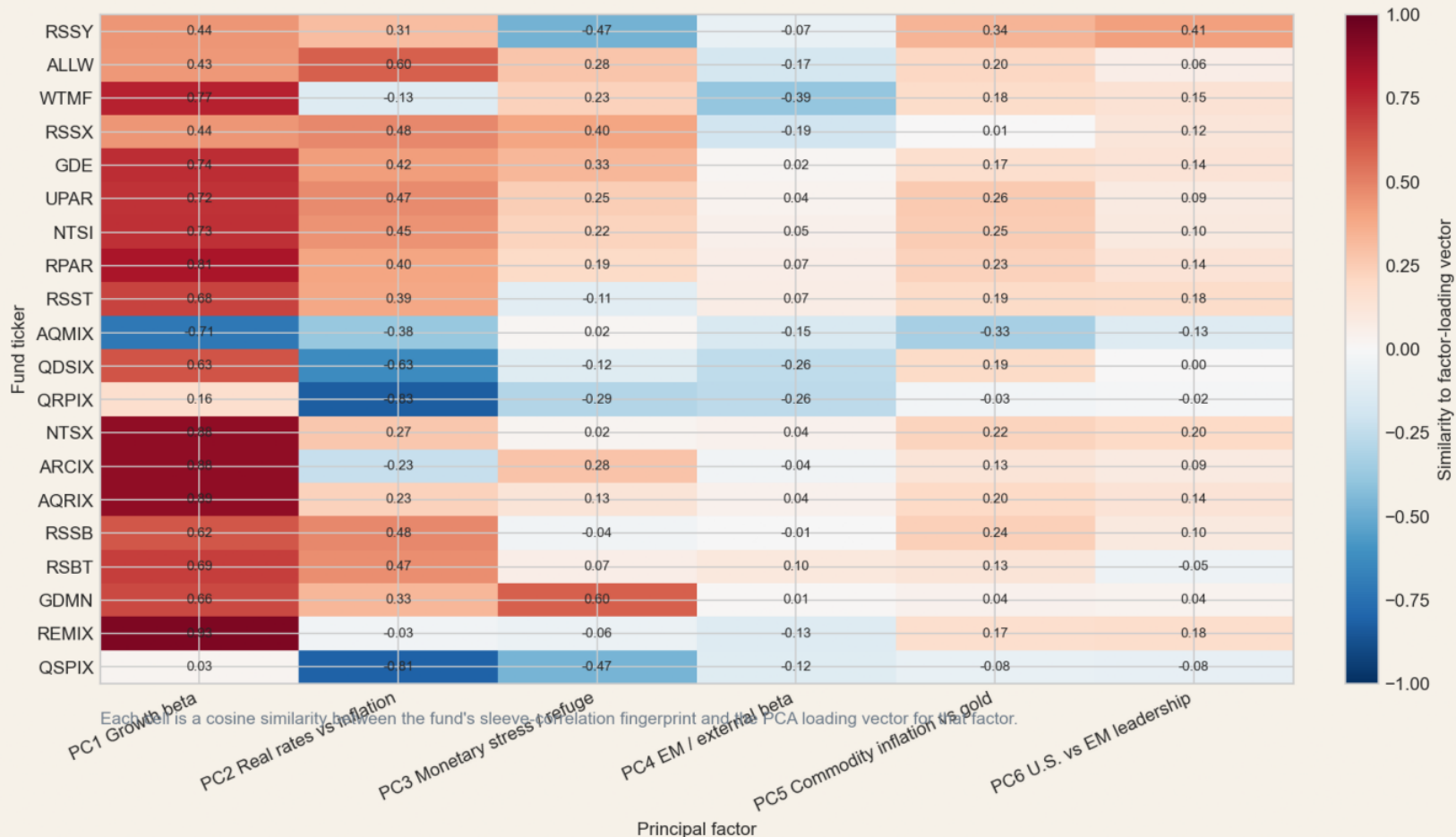


Beta Funds vs Core Beta Sleeves: Pairwise Daily Return Correlations

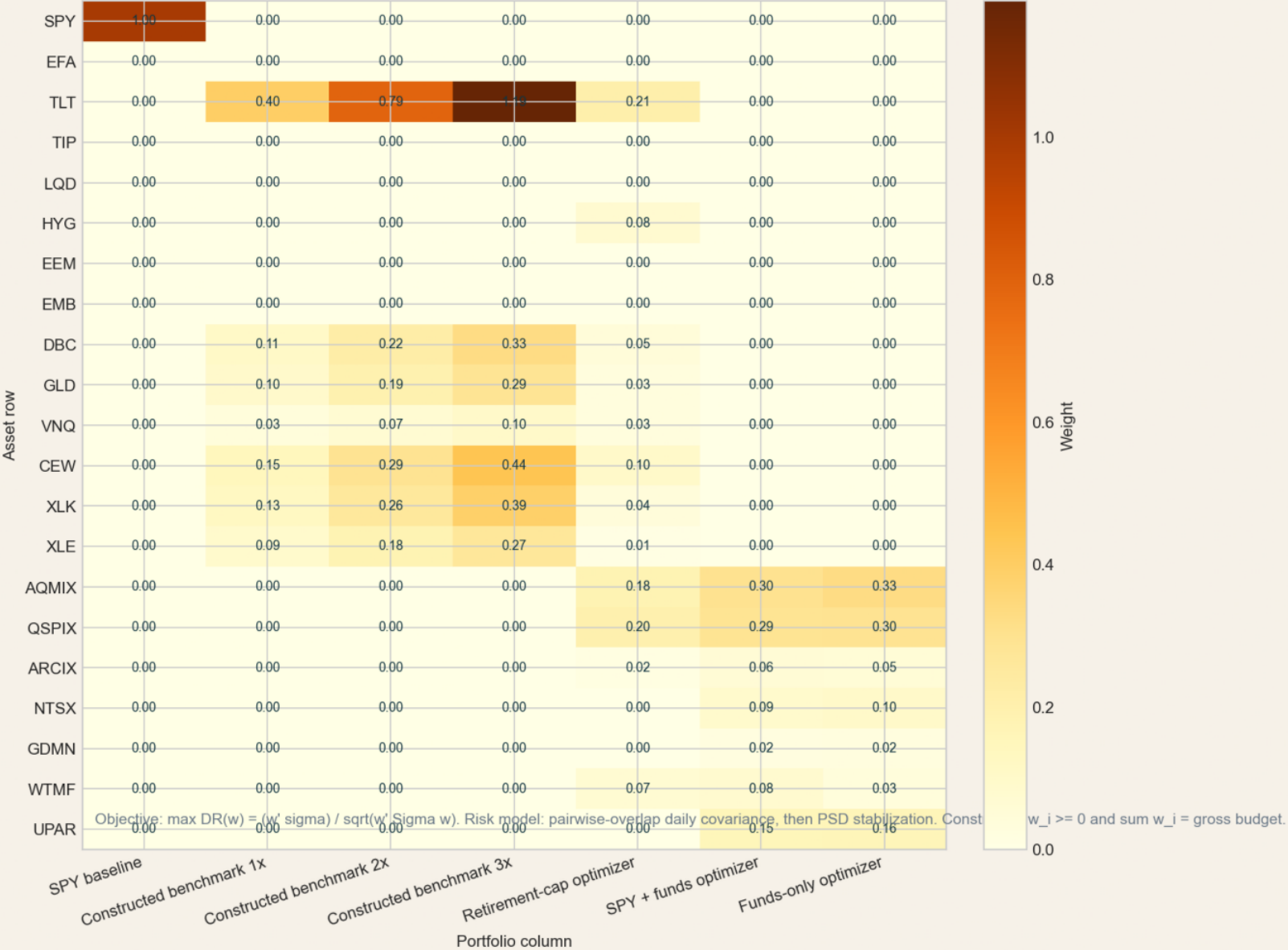


Fund-to-benchmark correlations use pairwise daily-return overlap, so short-history products like ALLW do not truncate the rest of the matrix even after the sleeve universe expands.

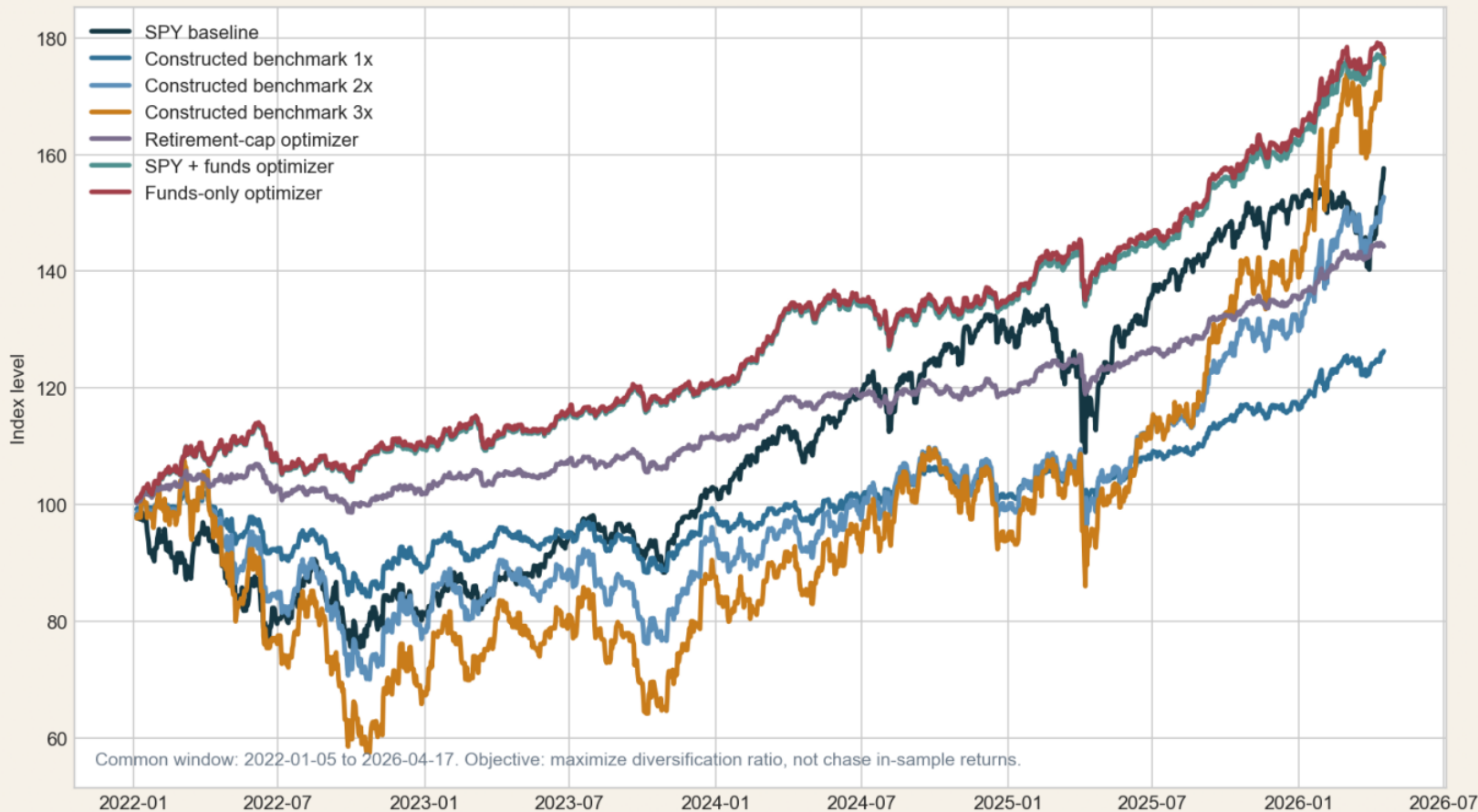
Beta Funds vs Macro Principal Factors



Portfolio Weight Matrix



Constructed Sleeves vs Accessible Alternatives



Constructed Benchmark 3x vs Flagship Diversified Funds

